Optimal Climate Policy with Incomplete Markets

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What we ask

Introduction

- Carbon emissions, the main source of global warming, are the classic example of an externality
- With no other distortions, the carbon tax equals the social cost of carbon: the **Pigouvian tax**
- With incomplete markets (inequality, risk) and distortionary taxes, the carbon tax interacts with other wedges

Should the carbon tax respond to inequality, risk, and fiscal constraints?

What we do

Introduction

Develop a climate model (Barrage, 2020) with the standard incomplete-markets model, SIM (Aiyagari, 1994), and solve its Ramsey problem:

1. **Theory:**

   → Characterize the optimal carbon tax with distorted intra- and intertemporal margins; compare it with the Pigouvian rule.

2. **Quantitative Model:**

   → Calibrate to US inequality, income risk and fiscal policy.
   → Solve for the optimal carbon tax along the transition under fiscal regimes of increasing flexibility: debt, transfers, income taxes.

3. **Link to policy:**

   → Measure the welfare gains of the optimal policy relative to IPCC's high-emissions scenario, and the cost of deviating from it.

What we find

Introduction

1. **Theory:**

   → Distortions of intra and intertemporal margins **introduce a wedge** between Pigouvian rule and optimal carbon tax.

2. **Quantitative Model:**

   → Optimal carbon tax **closely tracks social cost of carbon**.
   → The Pigou principle **is robust and holds regardless of fiscal constraints**.
   → True even when other instruments move allocations.

3. **Link to policy:**

   → Relative to IPCC's high-emissions scenario the optimal policy leads to **4.46% welfare gains**.
   → **Asymmetry** of welfare effects: postponing costs more than over-taxing.

Where we fit

Introduction

- Optimal climate policy with complete markets (e.g., Golosov, Hassler, Krusell, and Tsyvinski, 2014; Bovenberg and de Mooij, 1994; Barrage, 2020; Douenne et al, 2023)
  → **Our paper**: introduce incomplete markets

- Distributional effects of climate policy (e.g., Känzig, 2023; Fried et al, 2018 and 2024; Benmir and Roman, 2022)
  → **Our paper**: study optimal policy, analyze the transition, and account for welfare benefits of mitigation

- Optimal fiscal policy with incomplete markets (e.g., Conesa, Kitao, and Krueger, 2009; Dyrda and Pedroni, 2023)
  → **Our paper**: introduce climate change and study climate policy

The Model

Environment

Model

- Discrete time t; population N_t, growth $n_{t+1} = N_{t+1}/N_t - 1$
- Sector TFP grows exogenously: $A_{i,t} = (1 + g_{A,t})A_{i,t-1}$ (economy on a balanced-growth path) ▸ Details
- Ex ante heterogeneous households, type i: initial productivity $e_{i0} \in E$ and assets $a_{i0} \in A_0$; fraction α_i ($\sum_i \alpha_i = 1$)
- Idiosyncratic productivity $e_{it} \in E$ evolves stochastically; history e_i^t with probability $\pi_{it}(e_i^t)$
- Households save only in a risk-free asset, subject to a borrowing constraint
- Two production sectors: a final-good sector and an energy sector that generates CO_2 emissions
- Government levies distortionary taxes, issues debt, and sets climate policy

Preferences and budget

Model

Life-time expected utility of a type-i household:

$$\sum_t \beta^t N_t \sum_{e_i^t} \pi_{it}(e_i^t) \left[u\big(c_{it}(e_i^t), h_{it}(e_i^t)\big) - v(Z_t) \right]$$

- $\beta \in (0,1)$: discount factor; c: consumption; h: labor
- $u(\cdot)$: per-period utility; $v(Z_t)$: utility damages from the climate Z_t (**increasing, convex**)

Per-capita assets evolve according to:

$$(1+n_{t+1})a_{it+1}(e_i^t) = (1+(1-\tau_{K,t})(r_t-\delta))a_{it}(e_i^{t-1}) + T_t + w_t(1-\tau_{H,t})e_{it}h_{it}(e_i^t) - c_{it}(e_i^t)$$

- w_t, r_t: before-tax wage and interest rate; $\tau_{K,t}, \tau_{H,t}$: capital / labor income tax; T_t: lump-sum transfer

Production: Final Goods

Model

- Technology:

$$Y_t = (1 - D(Z_t))A_{1,t}F(K_{1,t}, H_{1,t}, E_t)$$

  → Y : output
  → K : capital
  → H : labor
  → E : energy
  → $D(Z_t)$: production damages from environmental degradation Z (**increasing, convex**)

- Firm rents capital and hires labor and energy at factor prices r_t, w_t, and $p_{E,t}$, respectively

Production: Energy Sector

Model

- Technology:

$$E_t = A_{2,t} G(K_{2,t}, H_{2,t}),$$

  → $G(\cdot)$: constant-returns-to-scale technology
  → $K_{2,t}$ and $H_{2,t}$: capital and labor inputs fully mobile across sectors.
  → $A_{2,t}$ denotes total factor productivity in the energy sector

- Energy production generates industrial CO_2 emissions,

$$E_t^M = (1 - \mu_t) E_t,$$

  → μ_t: share of energy generated using clean technologies

- Profits:

$$\mathcal{P}_t = p_{E,t} E_t - \tau_{E,t}(1 - \mu_t) E_t - w_t H_{2,t} - r_t K_{2,t} - \Theta(\mu_t, E_t),$$

  → $\Theta(\mu_t, E_t)$: abating emissions incurs a cost
  → $\tau_{E,t}$: a tax on carbon emissions

Government and Climate

Competitive Equilibrium

Model

- Government budget:

$$G_t + N_t T_t + R_t B_t = \tau_{H,t} w_t H_t + \tau_{K,t}(r_t - \delta) K_t + \tau_{E,t} E_t^M + B_{t+1},$$

  → G_t: government spending; B_t: debt; R_t: gross interest rate on debt

- The climate variable Z_t:

$$Z_t = J(S_0, E_0^M, \ldots, E_t^M, \eta_0, \ldots, \eta_t).$$

  → $\{E_t^M\}$: history of endogenous emissions
  → $\{\eta_t\}$: other exogenous drivers
  → S_0: initial conditions

Theory

Ramsey problem

Theory

Utilitarian planner:

$$\mathcal{W}(\mathcal{T}) = \sum_i \alpha_i \sum_t \beta^t N_t \sum_{e_i^t} \pi_{it}(e_i^t) \left(u(c_{it}(e_i^t), h_{it}(e_i^t)) - v(Z_t) \right)$$

Given K_0, B_0, and the initial distribution over (a_{i0}, e_{i0}), for each policy $\mathcal{T} = \{\tau_{Kt}, \tau_{Ht}, \tau_{Et}, T_t\}$:

- $X_i(\mathcal{T}) = \{a_{it+1}, c_{it}, h_{it}\}$: **individual** allocations (by type and history)
- $X(\mathcal{T}) = \{C_t, H_t, K_{t+1}, E_t, \mu_t, Z_t, B_{t+1}\}$: **aggregate** allocations
- $P(\mathcal{T}) = \{R_t, w_t, r_t, p_{E,t}\}$: **prices**

The Ramsey planner **announces and commits** to policies at date 0 and solves:

$$\max_{\mathcal{T}} \ \mathcal{W}(\mathcal{T}) \qquad \text{s.t.} \quad \{X_i(\mathcal{T}), X(\mathcal{T}), P(\mathcal{T})\} \in CE(\mathcal{T})$$

- $CE(\mathcal{T})$: competitive equilibrium allocations and prices induced by policy $\mathcal{T}$

Pigouvian Tax

Underlying Problem

Theory

The **Pigouvian tax** is defined as

$$\tau_{E,t}^{Pigou} = \frac{1}{\mathcal{W}_{c,t}} \sum_{j=0}^{\infty} \beta^j \left(\mathcal{W}_{c,t+j} D'_{t+j} A_{1,t+j} F_{t+j} - N_{t+j} \mathcal{W}_{Z,t+j} \right) J_{E_t^M, t+j},$$

where

- $\mathcal{W}_{c,t} = \sum_i \alpha_i \sum_{e_i^t} \pi_{it} u_{cit}$: the marginal utility of consumption, averaged across household types i and histories e_i^t
- $\mathcal{W}_{Z,t} = -v'(Z_t)$: the marginal utility damage from a higher temperature

Interpretation: the welfare cost of an extra CO_2 emitted at t, in units of date-t consumption, evaluated at a competitive-equilibrium allocation

Optimal Carbon Tax

Theory

Proposition

The Ramsey carbon tax satisfies a modified Pigouvian rule:

$$\tau_{E,t}^{SB} = \frac{1}{\nu_t} \sum_{j=0}^{\infty} \beta^j \left(\nu_{t+j} D'_{t+j} A_{1,t+j} F_{t+j} - N_{t+j} \mathcal{W}_{Z,t+j} \right) J_{E_t^M,t+j},$$

where ν_t is the multiplier on the resource constraint for the final consumption good satisfying:

$$\nu_t = \mathcal{W}_{c,t} + \sum_i \alpha_i \sum_{e_i^t} \pi_{it} (SD_{it} + LD_{it}),$$

with

$$SD_{i,t} \equiv u_{cc,it} \left(\kappa_{it}^s - \frac{R_t}{1+n_t} \kappa_{it-1}^s \right), \qquad LD_{i,t} \equiv \kappa_{it}^\ell \left(u_{cc,it} w_t (1 - \tau_{H,t}) e_{it} + u_{ch,it} \right),$$

and where κ^s and κ^ℓ are the multipliers on the modified Euler equation and labor-supply FOC.

Optimal Carbon Tax

Theory

- Assuming $\{\tau_{H,t}, \tau_{K,t}, T_t\}_{t=0}^{\infty}$ are optimized:

$$\nu_t = \mathcal{W}_{c,t} + \underbrace{\sum_i \alpha_i \sum_{e_i^t} \pi_{it}(SD_{it} + LD_{it})}_{\text{impact of distortionary taxes}},$$

where

$$SD_{it} \equiv u_{cc,it}\left(\kappa_{it}^s - \tfrac{R_t}{1+n_t}\,\kappa_{it-1}^s\right) \qquad \text{(Savings Distortions)}$$

$$LD_{it} \equiv \kappa_{it}^{\ell}\left(u_{cc,it}\, w_t(1-\tau_{H,t})\, e_{it} + u_{ch,it}\right) \qquad \text{(Labor Distortions)}$$

$\rightarrow$ $SD_{it} = 0$ if household is borrowing constrained in consecutive periods
$\rightarrow$ $LD_{it} = 0$ if there are no income effects on labor supply

- Constraints on $\{\tau_{H,t}, \tau_{K,t}, T_t\}_{t=0}^{\infty}$ imply **additional terms**.

Quantitative Strategy

Quantitative Strategy

Overview

(1) Taking the model to the data

- **Set by matching moments**: 26 parameters (preferences, technology, income process) chosen jointly to match 26 moments
  - → macro statistics: capital–output ratio, hours, CO_2 emissions, Frisch elasticity
  - → cross-section: distributions of wealth, earnings, income and hours; business owners
  - → earnings growth: dispersion, skewness and kurtosis of one-year changes
- **Set exogenously**: structural parameters from the literature, 2020 initial conditions, exogenous trends (DICE 2023), and the initial fiscal policy

(2) Solving the Ramsey problem numerically

- Start from the initial balanced growth path (BGP) and solve forward for the optimal transition until the economy settles in the final BGP

Households

Calibration

Per-period utility, including climate damages:

$$u(c, h, Z) = \frac{\left(c^{\gamma}(1-\varsigma h)^{1-\gamma}\right)^{1-\sigma} + \left(1+\alpha_z Z^2\right)^{\sigma-1}}{1-\sigma}$$

- **Preferences**: discount factor β, consumption weight γ, hours scale ς; curvature σ set by an elasticity of intertemporal substitution of 2/3
- **Frisch elasticity** (targeted; mean over employed households): $\varepsilon_F = \frac{1-\gamma(1-\sigma)}{\sigma}\frac{1-\varsigma h}{\varsigma h}$
- **Climate damages in utility**: α_z puts 26% of damages on utility (climate module)
- **Labor productivity** $z_t = s_t\,(1+e_t)$: persistent s_t (Markov) times transitory e_t (i.i.d.) ▸ Income process
- **Borrowing limit** $\underline{a}$

Technology

Backstop Price

Calibration

- **Final good**: capital share $\alpha = 0.36$, energy share $\nu = 0.04$; level A_1 set by 2020 world output ($83.5 trillion)

$$Y_t = \left(1 - D(Z_t)\right) A_{1,t} K_{1,t}^{\alpha} H_{1,t}^{1-\alpha-\nu} E_t^{\nu}$$

- **Energy**: capital share $\alpha_E = 0.60$; relative productivity A_2/A_1

$$E_t = A_{2,t} K_{2,t}^{\alpha_E} H_{2,t}^{1-\alpha_E}, \qquad E_t^M = (1 - \mu_t)\, E_t$$

- **Abatement**: cost exponent $c_2 = 2.6$; backstop price P_t^{back}, the cost of abating the last ton: $696 in 2020, falling 1% a year to 2050, 0.1% after

$$\Theta(\mu_t, E_t) = P_t^{back} \frac{\mu_t^{c_2}}{c_2} E_t$$

- **Capital**: depreciation $\delta = 0.09$; **growth** (DICE 2023): TFP growth, 1.59% in 2020 and declining; population path

Initial Fiscal Policy

Calibration

	Value	Source
Labor income tax τ_H	27.7%	Trabandt and Uhlig (2011), updated; 2015–2019
Capital income tax τ_K	33.6%	Trabandt and Uhlig (2011), updated; 2015–2019
Consumption tax τ_C	4.2%	Trabandt and Uhlig (2011), updated; 2015–2019
Carbon tax τ_E	\$6/tCO$_2$	Barrage and Nordhaus (2023)
Energy tax τ_I	0	Barrage and Nordhaus (2023)
Government debt / GDP	103%	FRED; 2015–2019
Transfers / GDP	2.22%	NIPA 3.12, working-age benefits excl. Medicaid; 2015–2019

- Government consumption is the residual of the budget.
- **Climate module** (Dietz and Venmans 2019; IPCC 2021): temperature linear in cumulative emissions, 0.45 °C per 1,000 GtCO$_2$; damages 2% of GDP at 2 °C, 74% in production, the rest in utility ▸ Climate Module

Calibration: Targeted Moments

Calibration

Macro aggregates	Data	Model
Capital–output ratio	2.57	2.50
Mean hours (share of time)	0.31	0.30
CO_2 emissions, 2020 (Gt)	38.2	37.2
Frisch elasticity	1.00	1.04

Panel: 1-year earnings growth	Data	Model
Variance	2.33	2.35
Kelley skewness	−0.12	−0.12
Moors kurtosis	2.65	2.62

Cross-section	Data	Model
Wealth share, bottom 20% (%)	−0.7	−0.5
Wealth share, 60–80th (%)	7.7	6.2
Wealth share, top 20% (%)	89.8	89.3
Wealth share, top 5% (%)	68.7	70.0
Wealth Gini	0.88	0.86
Net worth ≤ 0 (% of households)	12.9	14.7
Earnings share, bottom 20% (%)	2.3	3.2
Earnings share, 60–80th (%)	20.5	19.8
Earnings share, top 20% (%)	57.1	59.3
Earnings share, top 5% (%)	30.1	32.6
Earnings Gini	0.54	0.55
Earnings Gini, non-owners	0.50	0.52
Income Gini	0.56	0.57
corr(earnings, wealth)	0.53	0.52
Hours share, 20–40th (%)	15.4	17.8
Hours share, top 20% (%)	30.0	27.3
Business owners: households (%)	8.5	8.4
Business owners: wealth (%)	47.0	47.8
Business owners: earnings (%)	15.4	17.2

Data: SCF 2019, households with a head under 65 and not disabled; CPS 2018 (hours); Pruitt and Turner 2020 (growth); NIPA 2015–19 (K/Y); Global Carbon Budget 2022 (CO_2).

Calibration: Lorenz Curves

Calibration



Close fit for wealth, earnings and income; hours less dispersed (their Gini is not targeted).

Numerical method: parameterized instruments

Numerical Solution

Regime 1: fixed B/Y

parameters (5): $c_0, \ldots, c_4$

- **Carbon tax:** $\tau_{E,t} = \max\left\{0,\ \sum_{j=0}^{4} c_j\, P_j(t)\right\}$ for $t \le t_F$, constant after

  P_j: Chebyshev polynomials on $[0, t_F]$, $t_F = 150$ years; capped at the backstop price

- **Lump-sum transfers:** $T_t = T_t^{bal}$

  T_t^{bal}: the transfer that balances the budget with debt at $B_t/Y_t = 1.03$

- **Capital and labor income taxes:** $\tau_{K,t} = 33.6\%,\ \tau_{H,t} = 27.7\%$

  held at the benchmark rates

Numerical method: parameterized instruments

Numerical Solution

Regime 2: flexible B/Y

parameters (9): $c_0, \ldots, c_4,\ \Delta_1, \lambda_1, \Delta_2, \lambda_2$

- **Carbon tax:** $\tau_{E,t} = \max\left\{0,\ \sum_{j=0}^{4} c_j\, P_j(t)\right\}$ for $t \le t_F,$ constant after

  P_j: Chebyshev polynomials on $[0, t_F]$, $t_F = 150$ years; capped at the backstop price

- **Lump-sum transfers:** $T_t = T_t^{bal} + \Delta_1 e^{-\lambda_1 t} + \Delta_2 e^{-\lambda_2 t} + D,\ \ T_t \ge 0$

  T_t^{bal}: budget balance at $B/Y = 1.03$; D makes B/Y stationary from year 200; $\Delta_1 = \Delta_2 = 0$: regime 1

- **Capital and labor income taxes:** $\tau_{K,t} = 33.6\%,\ \tau_{H,t} = 27.7\%$

  held at the benchmark rates

Numerical method: parameterized instruments

Numerical Solution

Regime 3: flexible B/Y, τ_K, τ_H

parameters (11): $c_0, \ldots, c_4,\ \Delta_1, \lambda_1, \Delta_2, \lambda_2,\ \bar{\tau}_K, \bar{\tau}_H$

- **Carbon tax:** $\tau_{E,t} = \max\left\{0,\ \sum_{j=0}^{4} c_j\, P_j(t)\right\}$ for $t \le t_F$, constant after

  P_j: Chebyshev polynomials on $[0, t_F]$, $t_F = 150$ years; capped at the backstop price

- **Lump-sum transfers:** $T_t = T_t^{bal} + \Delta_1 e^{-\lambda_1 t} + \Delta_2 e^{-\lambda_2 t} + D,\ \ T_t \ge 0$

  T_t^{bal}: budget balance at $B/Y = 1.03$; D makes B/Y stationary from year 200; $\Delta_1 = \Delta_2 = 0$: regime 1

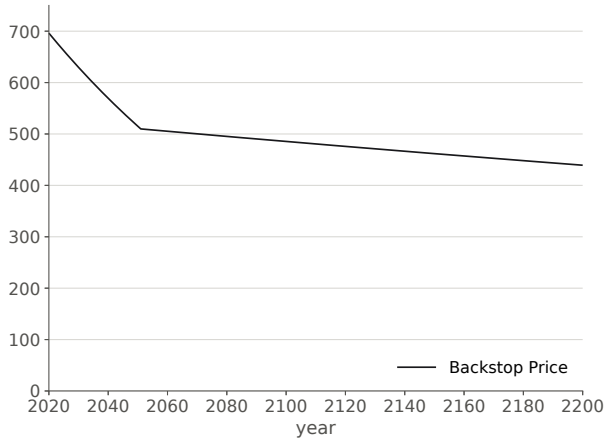
- **Capital and labor income taxes:** $\tau_{K,t} = \bar{\tau}_K,\ \tau_{H,t} = \bar{\tau}_H$ for all t

  revenue changes rebated through T_t^{bal}; the benchmark rates give regime 2

Results

Backstop Price (in $/tCO$_2$) Details

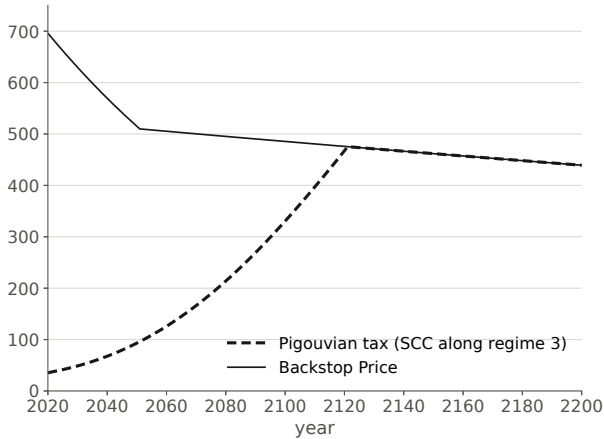
Results



Backstop price: the cost of eliminating the last ton of CO_2 (DICE 2023). At this price abatement is complete $\Rightarrow$ it caps the carbon tax.

Pigouvian Tax and Backstop Price (in $/tCO$_2$)

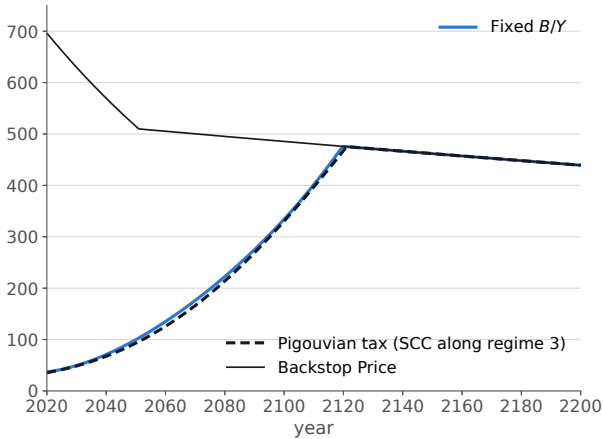
Results



Convex damages, cumulative temperature, growing output: each ton costs more than the last ⇒ the Pigouvian tax rises convexly, reaching the backstop in 2122.

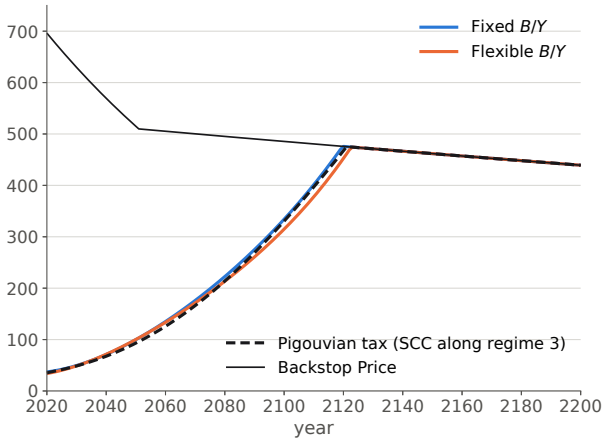
Optimal Carbon Taxes and Backstop Price (1/3)

Results



Optimal Carbon Taxes and Backstop Price (2/3)

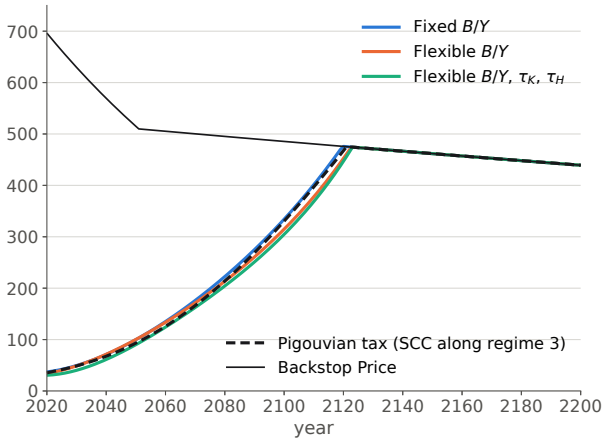
Results



Flexible debt front-loads transfers (see Aggregates); the carbon tax barely moves, within 8% of fixed B/Y.

Optimal Carbon Taxes and Backstop Price (3/3) Decomposition

Results



However the fiscal side is relaxed (debt, transfers, $\tau_K \to 44.4\%$, $\tau_H \to 35.2\%$), the three optimal taxes stay within 10% of the Pigouvian tax from 2040 on.

Why does the optimal tax track its own SCC?

Results

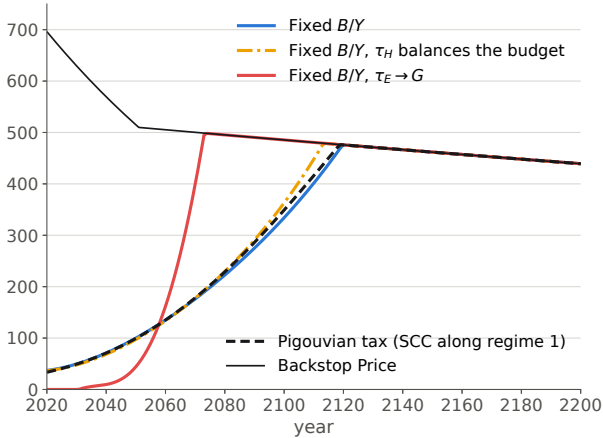
- **At its own SCC the climate margin is balanced**: marginal abatement cost = discounted marginal damage

$$\frac{\Theta_{\mu,t}}{E_t} \;=\; \tau_{E,t} \;=\; \mathrm{SCC}_t$$

- **A deviation can then pay only through the fiscal side**: it reaches households via $\{\, w_t(1-\tau_{H,t}),\ (1-\tau_{K,t})(r_t-\delta),\ T_t \,\}$
- **Unrestricted instruments** already set these margins optimally ⇒ no first-order gain left for the carbon tax (envelope)
- **Restricted instruments**: gain = shadow value of the instrument × quality of mimicry; factor prices barely respond to the carbon tax (energy's cost share 4%) ⇒ poor mimicry ▸ Magnitudes
- **Welfare is extremely flat around the optimum** ⇒ deviations from Pigou are worth very little. Waste the revenue and the fiscal flow itself is a first-order loss ⇒ the tax leaves the SCC (next)

Waste the revenue and the tax leaves the SCC

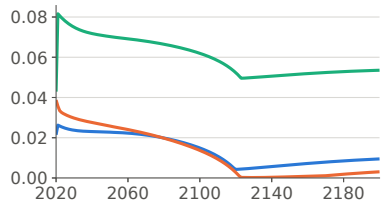
Results



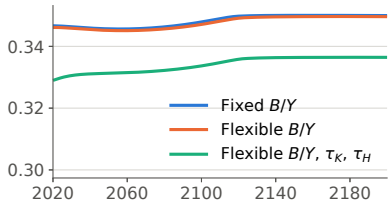
Not the closure per se: with τ_H balancing the budget the tax stays within 10% of Pigou. Wasting revenue costs most at intermediate rates $\Rightarrow$ near zero until 2040, at the backstop by 2074.

Aggregates

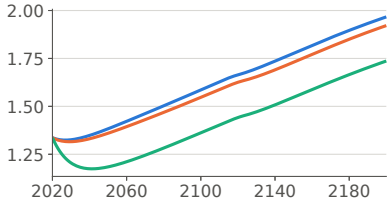
Results



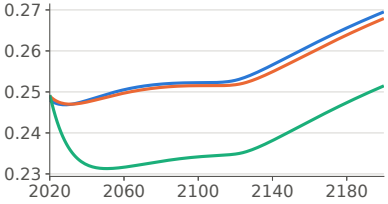
(a) Transfers to Output



(b) Effective Labor



(c) Capital



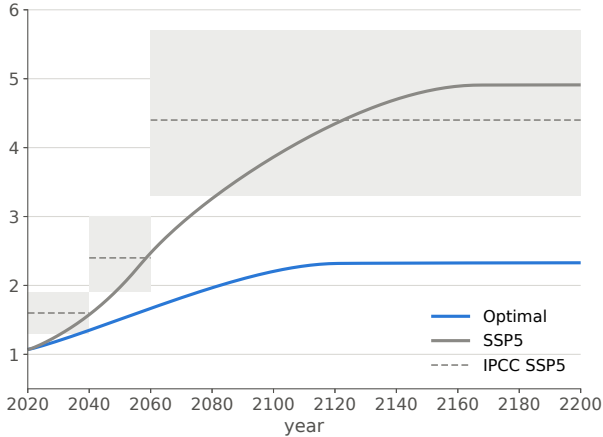
(d) Consumption

▸ Exogenous Growth Paths ▸ Why similar taxes?

Welfare Analysis

Shared Socioeconomic Pathway 5 (SSP5): Temperature

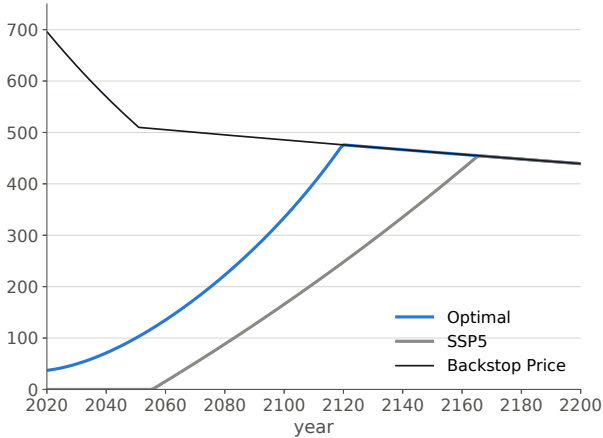
Welfare Analysis



SSP5: a late carbon-tax path; the model's temperatures sit at the low end of the IPCC's SSP5-8.5 ranges (3.6 vs 4.4 °C in 2081–2100).

Corresponding Carbon Taxes (in $/tCO$_2$)

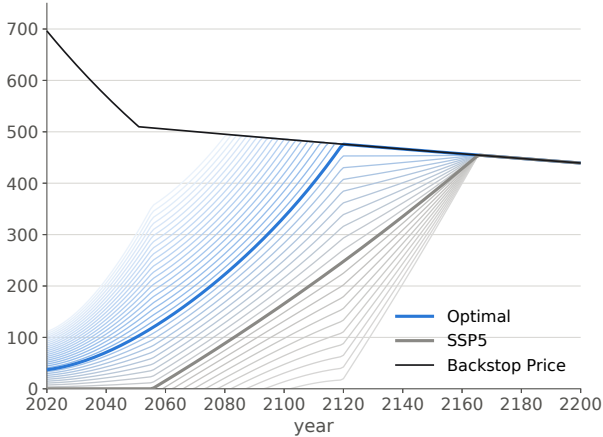
Welfare Analysis



Under SSP5 the carbon tax starts only in 2056 and lags the optimal path by 30 to 45 years.

Varying Taxes From One to the Other

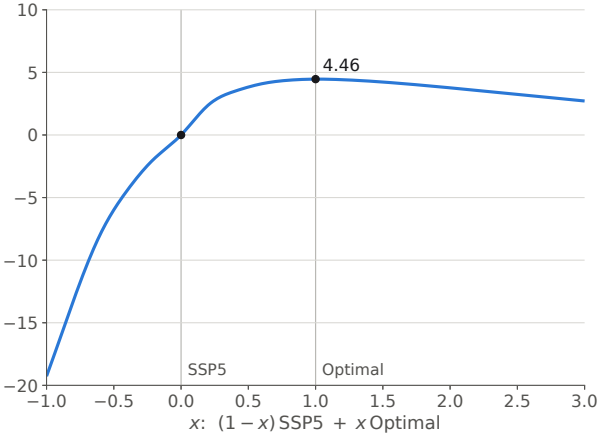
Welfare Analysis



Tax paths $(1 - x) \times$ SSP5 $+ x \times$ Optimal for x from -1 to 3, floored at zero and capped at the backstop.

Welfare Gain

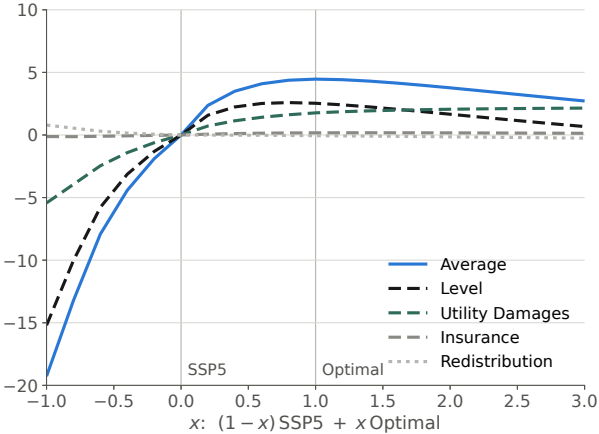
Welfare Analysis



Welfare peaks at the optimum (4.46%) and falls much faster below it: $x = 0$ loses 4.46 points, $x = 2$ only 0.69.

Welfare Decomposition

Welfare Analysis



Driven by efficiency (level, 2.5% at the optimum) and utility damages (1.8%); insurance and redistribution stay nearly flat.

Welfare across Regimes

Welfare Analysis

	CEV (% of consumption)	
	vs SSP5	vs Pigouvian tax
Fixed B/Y	4.46	0.002
Flexible B/Y	4.55	0.002
Flexible B/Y, τ_K, τ_H	5.34	0.006
Fixed B/Y, revenue wasted	2.70	1.219
Fixed B/Y, τ_H balances the budget	4.65	0.004

Flexibility of the fiscal instruments does its job: the welfare gain over SSP5 rises as instruments are added.

Welfare across Regimes

Welfare Analysis

	CEV (% of consumption)	
	vs SSP5	vs Pigouvian tax
Fixed B/Y	4.46	0.002
Flexible B/Y	4.55	0.002
Flexible B/Y, τ_K, τ_H	5.34	0.006
Fixed B/Y, revenue wasted	2.70	1.219
Fixed B/Y, τ_H balances the budget	4.65	0.004

Regardless of the regime, the Pigouvian principle holds strongly: the welfare gain from deviating from it is negligible.

Key takeaways

Conclusion

Three insights:

1. **Theory**: the optimal carbon tax deviates from Pigou, with a wedge driven by savings and labor distortions

2. **Quantitatively**: the Pigou principle is alive and robust in the SIM-climate model, regardless of fiscal constraints

3. **Policy**: better to overshoot the carbon tax than to undershoot; welfare is steep to the left of the optimum, flat to the right

Extensions:

- Stone-Geary extension (in the paper): energy as a necessity; results hold. ▸ Stone-Geary
- In progress: non-linear tax schedule, non-separable damages, state-contingent transfers.

Appendix

Where we fit (2)

Appendix

Other papers on optimal carbon taxes in an Aiyagari-type models:

- Bourany (2024): cross-country heterogeneity; no other fiscal instruments
- Kubler (2024): constrained-optimal policy à la Dávila et al. (2012); theory-focused, abstracts from redistribution
- Belfiori, Carroll, Hur (2024): constrained-optimal policy; combines theory and quantitative analysis
- Malafry and Brinca (2022): early attempt at full welfare maximization; stylized (2 periods, no other instruments)
- Wöhrmüller (2024): richer model and careful calibration, but fixes other instruments and focuses on steady-state

Our paper: solve for dynamic optimal fiscal policy to maximize social welfare, with a rich calibration and more flexibility over the instruments

Theory & Mechanisms

Pigouvian Tax: Underlying Problem

Back

Appendix

Planner chooses allocations subject only to technology, resources, and the climate module:

$$\max_{\{c_{it},\, h_{it},\, K_{t+1},\, E_t,\, \mu_t,\, Z_t\}} \sum_i \alpha_i \sum_t \beta^t N_t \sum_{e_i^t} \pi_{it}\big(u(c_{it}, h_{it}) - v(Z_t)\big)$$

s.t. the resource constraint (multiplier $\beta^t \nu_t$) and the climate module (multipliers $\beta^t \lambda_{Z,t}$)

Three first-order conditions:

- Consumption: $u_{c,it} = \nu_t$ for all i, $e_i^t \Rightarrow$ marginal utilities equalized: $\nu_t = \mathcal{W}_{c,t}$
- Temperature: $\lambda_{Z,t} = \nu_t\, D_t' A_{1,t} F_t - N_t \mathcal{W}_{Z,t}$
- Abatement: $\nu_t\, \theta_t'(\mu_t) = \sum_j \beta^j \lambda_{Z,t+j}\, J_{E_t^M, t+j}$, where $\Theta_t(\mu_t, E_t) = \theta_t(\mu_t) E_t$

Energy firms set $\theta_t'(\mu_t) = \tau_{E,t}$; combining the three conditions yields the Pigou formula

Ramsey problem: same conditions for Z_t and μ_t; household FOCs as constraints change the consumption condition: $\nu_t = \mathcal{W}_{c,t} + \sum_i \alpha_i \sum_{e_i^t} \pi_{it}(SD_{it} + LD_{it})$

Marginal cost of public funds

Back

Appendix

Corollary (link with the MCF). The carbon tax is the modified Pigouvian rule adjusted for the MCF, the ratio of the resource-constraint multiplier to the average marginal utility of consumption:

$$\text{MCF}_t \equiv \frac{\nu_t}{\mathcal{W}_{c,t}} = 1 + \frac{1}{\mathcal{W}_{c,t}} \sum_i \alpha_i \sum_{e_i^t} \pi_{it} \left(SD_{it} + LD_{it} \right)$$

- Deviations from Pigou arise whenever MCF $\neq 1$ (Barrage, 2020; Douenne et al., 2023)
- Complete markets: MCF averages to one under balanced growth $\Rightarrow$ Pigou on average (Douenne et al., 2023)
- Aiyagari economy: Ricardian equivalence breaks; MCF can lie above or below one
- Sign of $SD_{it} + LD_{it}$ is ambiguous: households may save or work too much or too little $\Rightarrow$ carbon tax above or below Pigou

Closeness to Pigou: Not by Construction

Back MCF

Appendix

- Evaluating both taxes at the same allocation fixes the units; the gap is governed by

$$\text{MCF}_t = \frac{\nu_t}{\mathcal{W}_{c,t}} = 1 + \frac{1}{\mathcal{W}_{c,t}} \sum_i \alpha_i \sum_{e_i^t} \pi_{it} \left(SD_{it} + LD_{it} \right)$$

- Nothing in the definition pins $SD_{it} + LD_{it}$: multipliers on household FOCs at the optimum
- The same convention delivers gaps elsewhere:
  - → Representative agent, constrained transfers: tax below Pigou (Bovenberg and Goulder, 1996; Barrage, 2020)
  - → Revenue thrown away ($\tau_E \to G$): optimal tax at zero until the 2030s, then at the backstop
- Allocation dependence of the SCC: 2 to 10% across evaluation points; figures hold the allocation fixed
- Welfare metric independent of the convention: imposing the own SCC costs 0.002 to 0.007 pp

Why similar taxes across scenarios? Back

Appendix

Optimal tax tracks the SCC; decompose it:

$$\tau_t^{e,Pigou} = \frac{1}{\lambda_t^A} \sum_{j=0}^{\infty} \beta^j \left(\lambda_{t+j}^A \, \lambda_{t+j}^B - \lambda_{t+j}^C \right) J_{E_t^M, t+j}$$

- λ^C (utility damages): pinned by temperature, similar across scenarios
- Debt and taxes crowd out capital, output falls:
  - $\rightarrow$ $\lambda^B = D' A_1 F$ falls (production damages proportional to output; 74% of damages)
  - $\rightarrow$ consumption falls, so $\lambda^A = \widetilde{\mathcal{W}}_c$ rises
- Damages enter as $\lambda^A \lambda^B$: fewer dollars of damage, each valued more
- $\sigma = 1.75$ (near log): offset close, not exact; SCC higher early, lower later

Competitive Equilibrium

Back

Appendix

Given K_0, B_0, a distribution of types i over (a_{i0}, e_{i0}), probabilities $\pi_{it}(e_i^t)$, populations $\{N_t\}_t$, and a tax policy $\mathcal{T} \equiv \{\tau_{H,t}, \tau_{K,t}, \tau_{I,t}, \tau_{E,t}, T_t\}_{t=0}^{\infty}$, a **competitive equilibrium** consists of individual variables $X_i \equiv \{a_{it+1}(e_i^t), c_{it}(e_i^t), h_{it}(e_i^t)\}_{i,t,e_i^t}$, aggregates $X \equiv \{C_t, H_t, H_{1,t}, H_{2,t}, K_{t+1}, K_{1,t}, K_{2,t}, E_t, \mu_t, Z_t, B_{t+1}\}_t$, and prices $P \equiv \{R_t, w_t, r_t, p_{E,t}\}_t$ such that: *(i)* households and firms optimize given prices and policies; *(ii)* the government satisfies its intertemporal budget constraint with bounded debt; *(iii)* temperature evolves according to (9); and *(iv)* markets clear: labor $H_t = H_{1,t} + H_{2,t}$ and capital $K_t = K_{1,t} + K_{2,t}$ split across sectors, and

$$C_t + G_t + K_{t+1} + \Theta_t(\mu_t)E_t = \big(1 - D(Z_t)\big)A_{1,t}F(K_{1,t}, H_{1,t}, E_t) + (1-\delta)K_t,$$

$$K_t + B_t = N_t \sum_i \alpha_i \sum_{e_i^{t-1}} \pi_{it-1}(e_i^{t-1}) a_{it}(e_i^{t-1}).$$

The economy is on a **balanced-growth path** if all aggregates except Z_t and μ_t grow at a constant rate and the equilibrium conditions hold.

How weak is the mimicry? Back of the envelope

Back

Appendix

- **Factor prices to the energy price:** substituting E out of F (Cobb-Douglas, K_1, H_1 fixed)

$$\frac{d\ln w}{d\ln p_E} \;=\; \frac{d\ln r}{d\ln p_E} \;=\; -\frac{\nu}{1-\nu} \;=\; -0.042$$

  $\rightarrow$ w and r move equiproportionally: the bundle has one direction

- **Energy price to the tax:** $p_E = mc_E + \tau_E(1-\mu) + \Theta/E$ and $\Theta_\mu = \tau_E E$

$$\frac{d\ln p_E}{d\ln \tau_E} \;=\; \frac{\tau_E(1-\mu)}{p_E} \;\equiv\; s_\tau \;\approx\; 0.26 \;\; (2020); \qquad s_\tau \to 0 \text{ at the backstop}$$

- **Product:** $\dfrac{d\ln w}{d\ln \tau_E} = -\dfrac{\nu}{1-\nu}\, s_\tau \approx -0.011$: doubling the tax moves w, r by 1%
- **Against the target:** freed, τ_H goes 27.7 to 35.2%: an 11 log-point after-tax wage move
  $\rightarrow$ requires $\Delta \ln \tau_E \approx 10$; room to the backstop: 2.6
  $\rightarrow$ revenue ceiling $s_\tau\, \nu \approx 1\%$ of output; transfer expansion when freed: 2 to 5.5pp of GDP

One direction, a tenth of the required move, vanishing at the backstop.

Decomposing the SCC

Back

Appendix

Decompose the Pigouvian tax (i.e., the SCC) as follows:

$$\tau_t^{e,Pigou} = \frac{1}{\lambda_t^A} \sum_{j=0}^{\infty} \beta^j \left(\lambda_{t+j}^A \lambda_{t+j}^B - \lambda_{t+j}^C \right) J_{E_t^M, t+j}, \tag{1}$$

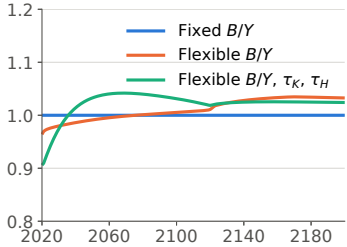
with

$$\lambda_t^A = \widetilde{\mathcal{W}}_{c,t} \quad ; \quad \lambda_t^B = D_t' A_{1,t} F_t \quad ; \quad \lambda_t^C = \mathcal{W}_{Z,t}. \tag{2}$$

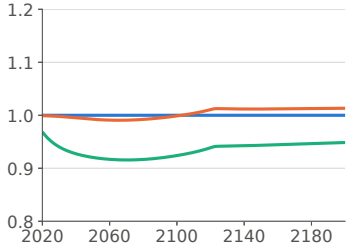
Components

Back

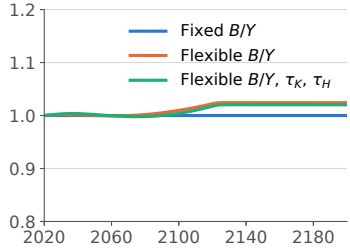
Appendix



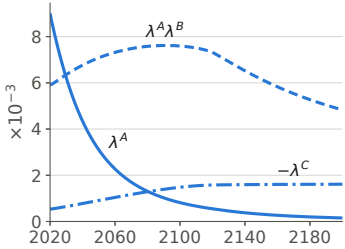
(e) Marginal Utility of Consumption



(f) Marginal Production Damages (λ_t^B)



(g) Marginal Utility from Climate (λ_t^C)



(h) Scale of Components

Welfare decomposition

Appendix

- The utilitarian welfare function can increase for four reasons:

  1. Reduction in distortions, if the utility of the average agent, $\sum_{t=0}^{\infty} \beta^t u(C_t, N_t)$, increases: **the level effect (Δ_L)**
  2. Lower utility damages from climate, $\sum_{t=0}^{\infty} \beta^t v(Z_t)$: **the utility-damages effect (Δ_{UD})**
  3. Transfers from ex-post rich to ex-post poor, if the risk of each individual path $\{c_t, n_t\}_{t=1}^{\infty}$ is reduced: **the insurance effect (Δ_I)**
  4. Transfers from ex-ante rich to ex-ante poor, if the inequality between certainty equivalents for $\{c_t, n_t\}_{t=1}^{\infty}$ is reduced: **the redistribution effect (Δ_R)**

Let Δ be the utilitarian (average) welfare gain, which decomposes as:

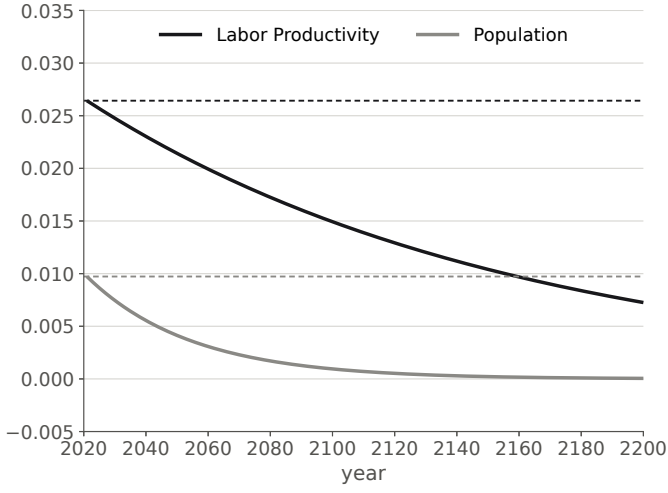
$$(1+\Delta) = (1+\Delta_L)\,(1+\Delta_{UD})\,(1+\Delta_I)\,(1+\Delta_R)$$

Additional Results

Productivity and population growth over transition

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Appendix



Distribution of Gains

Appendix



All households gain, 4.3 to 6.6%: flat except in the top wealth quintile, highest for its top earners.

Backstop Price of Carbon Abatement

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Appendix

- The **backstop price** is the marginal cost of abating the last unit of CO_2.
- At this price, technology is available to:
  - → Fully capture all emitted carbon, or
  - → Produce carbon-free energy at scale to fully substitute fossil fuels.
- Key question: What is the **cost per ton of CO_2** for the cheapest such technology?
- The DICE model uses an **exogenous path** for the backstop price.
- This path is taken from a synthesis of external studies.
- It is **independent of the model's internal calibration**.

Calibration

Climate Module

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Appendix

	Value	Source
Carbon cycle and temperature		
Cumulative emissions, 2020	2,390 $GtCO_2$	IPCC (2021)
Temperature, 2020 (above 1850–1900)	1.07 °C	IPCC (2021)
Temperature response to cumulative emissions	0.45 °C per 1,000 $GtCO_2$	IPCC (2021)
Pulse-adjustment rate	0.5	Dietz and Venmans (2019)
Land-use emissions, 2020	4.17 $GtCO_2$	Global Carbon Budget 2022
their decline	10% per 5 years	DICE 2023
Damages $D(Z) = 1 - e^{-\gamma_D Z^2/2}$		
Curvature γ_D	0.01	Dietz and Venmans (2019)
Share of damages in production (rest in utility)	74%	Barrage (2020)

Climate Module: Dietz and Venmans (2019) and DICE 2023

Calibration

Temperature dynamics

- Global mean temperature:

$$Z_{t+1} = Z_t + \epsilon\big(\zeta \mathcal{E}_t - Z_t\big)$$

$\zeta = 0.00045$°C/GtCO$_2$, $\epsilon = 0.5$

- Cumulative emissions:

$$\mathcal{E}_{t+1} = \mathcal{E}_t + E_t^M + E_t^{\text{ex}}$$

land emissions (DICE 2023) fall 10%/5yr

- Initial (IPCC 2021; GCP 2022): Z_{2020}=1.1, $\mathcal{E}$=2390, E^M=36, E^{ex}=4

Production damages

- Following Dietz and Venmans (2019):

$$D(Z_t) = 1 - \exp\big(-\tfrac{\alpha_1}{2} Z_t^2\big)$$

- Curve similar to DICE 2023
- $\alpha_1 = 0.01$: damages of 2% of GDP at 2°C, 7.7% at 4°C

Calibration: Population Partition

Appendix

	Share of the total (%)			
	Population	**Earnings**	**Income**	**Wealth**
Workers				
Data	91.5	84.6	78.9	53.0
Model	91.6	82.8	75.2	52.2
Business owners				
Data	8.5	15.4	21.1†	47.0
Model	8.4	17.2	24.8†	47.8

Households with a head under 65 and not disabled (SCF 2019). Owners' shares of population, earnings and wealth are targeted; † not targeted. Workers are all other households, including those with no labour income in the year (data 5.4%, model 2.4%).

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Income Process

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Appendix

Labor productivity $z_t = s_t\,(1 + e_t)$, normalized to mean one

- **Persistent** $s_t \in \{s_1, \ldots, s_4\}$, Markov chain: levels, exit probabilities and shares directing the exits
- **Transitory** e_t, i.i.d.: nodes $\{d_{lo}, 0, d_{hi}\}$ with probabilities π_1, π_2, π_3, each times a mean-one log-normal shock with spread σ_c
- **Business owners**: the wealthiest households in the two high states, one share per state; a label for measurement only

Population Partition

Exogenous growth and detrending

Back

Appendix

Exogenous trends.

- Population: $N_{t+1} = (1 + n_{t+1})N_t$
- Sector TFP: $A_{i,t} = (1 + g_{A,t})\, A_{i,t-1}, \quad i \in \{1, 2\}$, with $g_{A,t}$ declining ($g_{A,2020} = 0.0159$, DICE 2023)

Labor-augmenting representation. Final good $Y_t = A_{1,t} K_{1,t}^{\alpha} H_{1,t}^{1-\alpha-\nu} E_t^{\nu}$ is Cobb-Douglas, so TFP maps to a labor-augmenting trend:

$$\Psi_t \equiv A_{1,t}^{1/(1-\alpha-\nu)}, \qquad \Psi_{t+1} = (1 + g_{t+1})\Psi_t$$

Detrending. Stationary variables $\tilde{c}_t = c_t/\Psi_t$, $\tilde{a}_t = a_t/\Psi_t$, $\tilde{w}_t = w_t/\Psi_t$, $\tilde{T}_t = T_t/\Psi_t$:

- Borrowing limit grows with technology, $\underline{a}_{t+1} = \bar{a}\, \Psi_{t+1}$, so a balanced-growth path exists given policy
- Solve the stationary economy with time-invariant methods Numerical Method

Climate Module

Temperature

Appendix

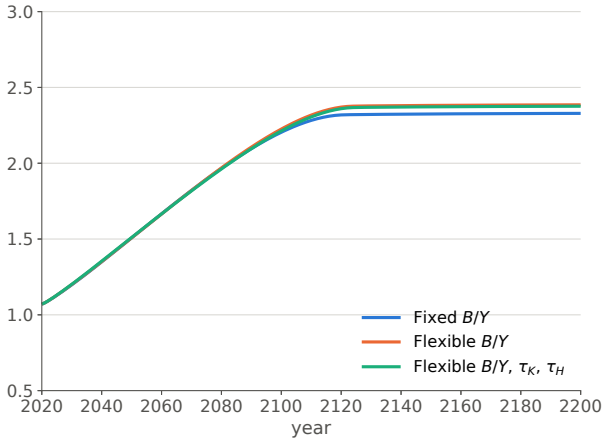
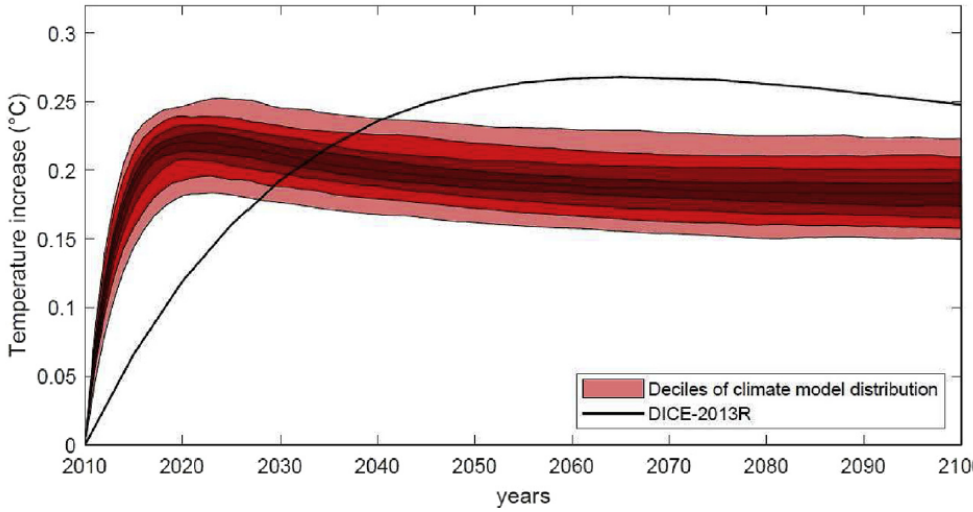


Figure: Temperature Change

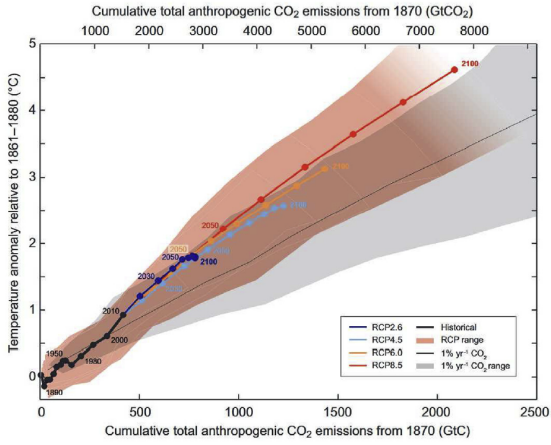
Temperature response to emissions

Appendix



Temperature linear in cumulative emissions

Appendix



Source: Dietz and Venmans (2019)

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Extension: Stone-Geary

Extension: Energy Good and Stone-Geary Preferences

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Appendix

- Poorer households emit less in total, but their consumption is more carbon-intensive (Sager, 2019): energy is a necessity
- Households consume the final good c and energy d directly:

$$u_t(c_{it}, d_{it}, h_{it}) = \frac{\left(c_{it}\left(d_{it} - \bar{d}_t\right)^{\epsilon}\left(1 - \varsigma h_{it}\right)^{\gamma}\right)^{1-\sigma}}{1-\sigma}$$

  → $\bar{d}_t = \bar{d}\,\Psi_t$: subsistence energy, grows with productivity (balanced growth)
  → Intratemporal condition: $u_d/u_c = p_{D,t} + \tau_{D,t}$, with $\tau_{D,t}$ a tax on household energy

- Energy market splits: $E_t = E_{1,t} + D_t$ (firms + households)
- Emission intensity ι of household energy: emissions $\propto E_{1,t} + \iota D_t$

Stone-Geary: Calibration

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Appendix

Choose $(\bar{d}, \epsilon)$ to match the distribution of household energy use (CEX, 2019):

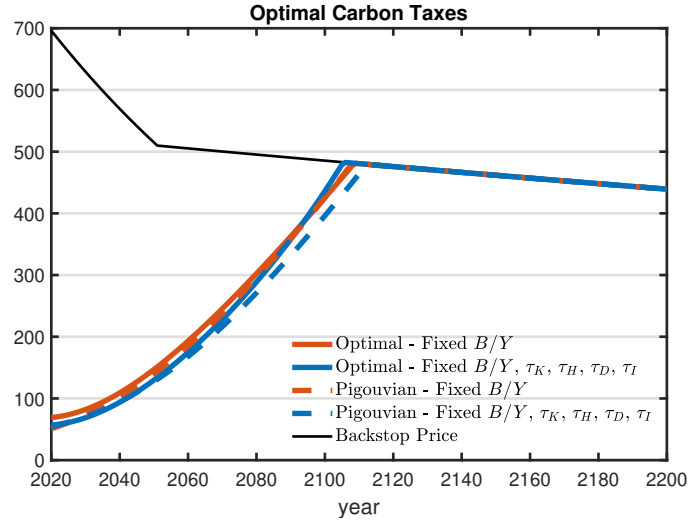
	Expenditure quintiles					Bottom/Top 5%	Gini
	1st	2nd	3rd	4th	5th		
Energy share (%)	11.3	16.6	20.0	23.3	28.9	1.8 / 8.5	0.178

- Household energy share of total expenditures: 7.0% (CEX, 2019)
- Household share of total emissions: 30% (EPA, 2024: residential + passenger vehicles)
- Remaining parameters: same procedure as the benchmark

Stone-Geary: Optimal Carbon Taxes (in $/tCO$_2$)

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Appendix



Optimal tax on the SCC in both scenarios; welfare cost of simply Pigou: 0.008% and 0.001%